

1. Preliminaries

Def: $(G, *)$ group if

- $g, h \in G \Rightarrow g * h \in G$
- $\exists e \in G : g * e = e * g = g \quad \forall g \in G$
- $\forall g \in G \exists g^{-1} \in G : g * g^{-1} = g^{-1} * g = e$
- $g * (h * k) = (h * k) * g \quad \forall g, h, k \in G$

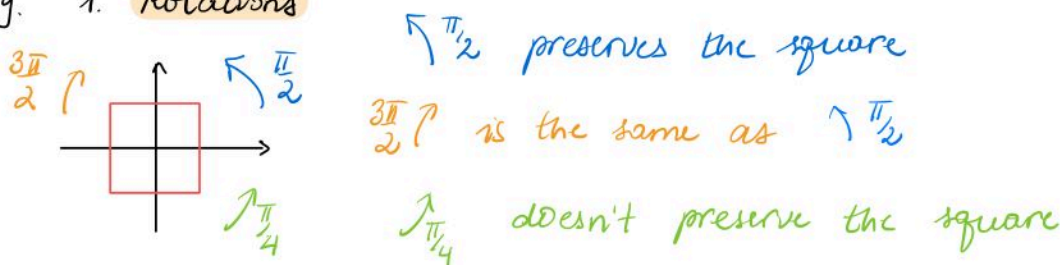
- Suppose $(G, +)$ is a group.
 $H \subseteq G$ is a subgroup of G if $(H, +)$ is a group.

Def: $T: X \rightarrow Y$ invertible if

$$\begin{aligned} \exists S: Y \rightarrow X : \quad & S(T(P)) = P \quad \forall P \in X \\ & T(S(Q)) = Q \quad \forall Q \in Y \\ T^{-1} := S & \text{ inverse of } T \end{aligned}$$

2. Symmetries : invertible functions from some set to itself preserving some feature of the set.

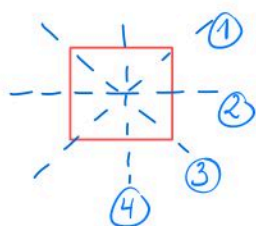
e.g. 1. Rotations



If $T := \mathcal{R}_\alpha$, then the inverse of T is $T^{-1} = \mathcal{R}_{-\alpha}$

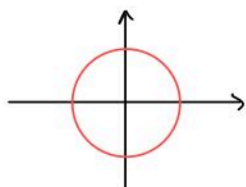
The square $\square$ has 4 rotation symmetries: 0° , 90° , 180° , 270°

2. Reflections



The square has 4 reflection symmetries

Another example:



The circle has "continuous" symmetry.

3. Groups of symmetries

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① Group of symmetries of $\mathbb{R}^n$

Def: $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ **vector space symmetry** of $\mathbb{R}^n$ if

- $\forall v, w \in \mathbb{R}^n \quad T(v+w) = T(v) + T(w)$
 $T(av) = aT(v)$
- T invertible

So T is an invertible linear transformation

Recall: • $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ linear transformation

$$\Rightarrow \exists M_T \in \mathcal{M}(n, \mathbb{R}) : T(v) = M_T v \quad \forall v \in \mathbb{R}^n$$

- Conversely, every $M \in \mathcal{M}(n, \mathbb{R})$ defines a lin. transformation

$$T_M: \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad v \mapsto Mv.$$

General linear group

$$GL(n, \mathbb{R}) = \{ M \in \mathcal{M}(n, \mathbb{R}) : \det M \neq 0 \}$$

$$\cong GL(\mathbb{R}^n) = \{ T: \mathbb{R}^n \rightarrow \mathbb{R}^n : T \text{ linear and invertible} \}$$

→ Claim: $(GL(n, \mathbb{R}), \cdot_{\mathcal{M}(n, \mathbb{R})})$ and $(GL(\mathbb{R}^n), \circ)$ are groups

Pf: Ex (use $\det(MN) = \det M \cdot \det N$)

Special linear group

$$SL(n, \mathbb{R}) = \{ M \in \mathcal{M}(n, \mathbb{R}) : \det M = 1 \}$$

$$\cong SL(\mathbb{R}^n) = \{ T: \mathbb{R}^n \rightarrow \mathbb{R}^n : \det M_T = 1 \}$$

→ Claim: - $SL(n, \mathbb{R})$ is a subgroup of $GL(n, \mathbb{R})$

- $SL(\mathbb{R}^n)$ is a subgroup of $GL(\mathbb{R}^n)$.

② Group of symmetries of Euclidean space $\mathbb{R}^2$ → **preserve Euclidean distance**

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$: - T invertible

- T linear transformation

- T **isometry**: $\forall P, Q \in \mathbb{R}^2 \quad \|T(P) - T(Q)\|_2 = \|P - Q\|_2$

(*)

Orthogonal group in Euclidean sense: $O(\mathbb{R}^2) := \{ T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ as in } (*) \}$

→ Claim: $(O(\mathbb{R}^2), \circ)$ is a group

Pf: Ex

reflection across a line
 through $(0,0)$ w/ angle $\alpha/2$
 with positive x -axis

$$\rightarrow \text{claim: } O(2, \mathbb{R}) = \left\{ \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix} : \alpha \in [0, 2\pi) \right\}$$

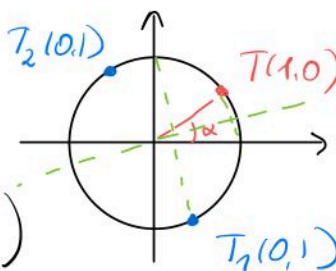
Pf: $\subseteq$: Let $M \in O(2, \mathbb{R}) := \{M \in M(2, \mathbb{R}) : T_M \in O(\mathbb{R}^2)\}$. Then

$$- \|T_M(1,0)\|_2 = \|(1,0) - (0,0)\|_2 = 1, \quad \|T(0,1)\|_2 = 1$$

$$- T(1,0) = (\cos \alpha, \sin \alpha), \quad \alpha \in [0, \pi/2]$$

$$- \|T(1,0) - T(0,1)\|_2 = \sqrt{2}$$

$$\Rightarrow T_1(0,1) = \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix} \quad \text{or} \quad T_2(0,1) = \begin{pmatrix} \sin \alpha \\ -\cos \alpha \end{pmatrix}$$



$\supseteq$: easy

□

$\rightarrow$ Claim: $O(2, \mathbb{R})$ is a group

Pf: Ex: $T_\alpha^{\text{rot}} \circ T_\beta^{\text{rot}} = T_{\alpha+\beta}^{\text{rot}} = T_\beta^{\text{rot}} \circ T_\alpha^{\text{rot}}$

$$T_\beta^{\text{rot}} \circ T_\alpha^{\text{ref}} = T_{\alpha+\beta/2}^{\text{ref}}$$

$$T_\alpha^{\text{ref}} \circ T_\beta^{\text{rot}} = T_{\alpha-\beta/2}^{\text{ref}}$$

$$T_\alpha^{\text{ref}} \circ T_\beta^{\text{ref}} = T_{\alpha-\beta}^{\text{rot}}$$

Remark: • set of all rotations is a group:

$SO(2, \mathbb{R})$ special orthogonal group

• set of reflections is NOT a group

③ Group of symmetries of Euclidean space $\mathbb{R}^3$ ($\mathbb{R}^n$)

Preserving the distance = Preserving the dot product

$$\|p\|_2 = \sqrt{p \cdot p}$$

$$\|T(v) - T(w)\|_2 = \|v - w\|_2 \Leftrightarrow T(v) \cdot T(w) = v \cdot w \quad \forall v, w \in \mathbb{R}^n$$

Pf: $\Leftarrow$: $\|T(v) - T(w)\|_2 \stackrel{T \text{ lin.}}{=} \|T(v-w)\|_2 = \sqrt{T(v-w) \cdot T(v-w)}$

$$= \sqrt{(v-w) \cdot (v-w)} = \|v-w\|_2$$

$$\Rightarrow \|T(v) - T(w)\|_2 = \|v-w\|_2 \Rightarrow (T(v) - T(w)) \cdot (T(v) - T(w)) = (v-w) \cdot (v-w)$$

$$\Rightarrow T(v) \cdot T(v) - 2T(w) \cdot T(v) + T(w) \cdot T(w) = v \cdot v - 2v \cdot w + w \cdot w$$

Take $w=0$: $\|T(v) - T(0)\|_2 = \|T(v)\|_2 = \|v-0\|_2 = \|v\|_2 \quad \forall v \in \mathbb{R}^n$

$$\Rightarrow \|v\|_2^2 - 2T(w) \cdot T(v) + \|w\|_2^2 = \|v\|_2^2 - 2v \cdot w + \|w\|_2^2$$

$$\Rightarrow T(w) \cdot T(v) = w \cdot v \quad \forall w, v \in \mathbb{R}^n$$

□

$O(\mathbb{R}^3) = \{ T \in GL(\mathbb{R}^3) : T \text{ preserves the dot product} \}$ (Euclidean orthogonal group) 4

→ Claim: $O(\mathbb{R}^3)$ is a group

Pf:

- $I \in O(\mathbb{R}^3)$
- $S, T \in O(\mathbb{R}^3) \Rightarrow T \circ S \in O(\mathbb{R}^3)$, since

$$(T \circ S)(v) \cdot (T \circ S)(w) = T(S(v)) \cdot T(S(w)) = S(v) \cdot S(w) = v \cdot w$$
- $T \in O(\mathbb{R}^3) \Rightarrow T^{-1} \in O(\mathbb{R}^3)$, since

$$\forall x, y \in \mathbb{R}^3 \exists v, w \in \mathbb{R}^3 : T(v) = x, T(w) = y$$

$$\Rightarrow T^{-1}(x) \cdot T^{-1}(y) = v \cdot w = T(v) \cdot T(w) = x \cdot y$$

□

$SO(\mathbb{R}^3) = \{ T \in O(\mathbb{R}^3) : \det(T) = 1 \}$ rotation group / special orthogonal group

→ subgroup of $O(\mathbb{R}^3)$

Q: What is the matrix group corresponding to $O(\mathbb{R}^3)$?

A: $O(3, \mathbb{R}) = \{ M \in \mathcal{M}(3, \mathbb{R}) : M^T M = I_3 \}$

→ Pf: Let $S: \mathbb{R}^3 \rightarrow \mathbb{R}^3 : S(v) = Mv \quad \forall v \in \mathbb{R}^3$.

$$S(v) \cdot S(w) = (Mv) \cdot (Mw) = (Mv)^T (Mw) = v^T M^T M w = v^T (M^T M) w \quad (*)$$

Therefore $S \in O(\mathbb{R}^3) \Leftrightarrow M^T M = I_3$ (take $v = e_j, w = e_i$ in $(*)$)

□

→ Claim: $O(3, \mathbb{R})$ is a group

Pf: - $I_3 \in O(3, \mathbb{R})$

$$- M_1, M_2 \in O(3, \mathbb{R}) \Rightarrow (M_1 M_2)^T (M_1 M_2) = M_2^T (M_1^T M_1) M_2 = M_2^T M_2 = I_3$$

$$\Rightarrow M_1 M_2 \in O(3, \mathbb{R})$$

$$- M \in O(3, \mathbb{R}) \Rightarrow M^T M = I_3 \Rightarrow \det M^T \cdot \det M = 1 \Rightarrow (\det M)^2 = 1$$

$$\Rightarrow \det M = \pm 1 \neq 0$$

$$\Rightarrow \exists M^{-1} : M \cdot M^{-1} = I_3$$

$$\Rightarrow M^T M \cdot M^{-1} = I_3 \cdot M^{-1} \Rightarrow M^T = M^{-1}$$

$$\Rightarrow (M^{-1})^T M^{-1} \stackrel{M^T = M^{-1}}{=} (M^T)^T M^{-1} = M \cdot M^{-1} = I_3$$

□

Remark: $O(n, \mathbb{R}) = \{ M \in \mathcal{M}(n, \mathbb{R}) : M^T M = I_n \}$ orthogonal group of degree n over real numbers

$SO(n, \mathbb{R}) = \{ M \in O(n, \mathbb{R}) : \det M = 1 \}$ special orthogonal group of degree n over real numbers

④ Group of symmetries of Lorentz plane $\mathbb{R}^2$ (spacetime)

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Def. & Thm: Lorentz product on $\mathbb{R}^2$: $(t, x_1) \cdot (t, x_2) := t_1 t_2 - x_1 x_2$

has the following properties:

- symmetric
- bilinear
- nondegenerate

Pf: Ex

Lorentz group: $\mathcal{L}(\mathbb{R}^2) = \{T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 : T \text{ linear}, T(v) \cdot T(w) = v \cdot w \ \forall v, w \in \mathbb{R}^2\}$

→ Claim: $(\mathcal{L}(\mathbb{R}^2), \circ)$ is a group

Pf: - $I_2 \in \mathcal{L}(\mathbb{R}^2)$

- $S, T \in \mathcal{L}(\mathbb{R}^2) \Rightarrow T \circ S \in \mathcal{L}(\mathbb{R}^2)$

- $T \in \mathcal{L}(\mathbb{R}^2), T(v) = 0 \Rightarrow v \cdot w = T(v) \cdot T(w) = 0 \ \forall w \in \mathbb{R}^2$
 $\Rightarrow v = 0$
 $\Rightarrow T$ invertible

Let $v, w \in \mathbb{R}^2$. Then $\exists x, y \in \mathbb{R}^2$ s.t. $T(x) = v, T(y) = w$. Then

$$T^{-1}(v) \cdot T^{-1}(w) = x \cdot y = T(x) \cdot T(y) = v \cdot w$$

□

Group of Lorentz rotations $\mathcal{L}^+(\mathbb{R}^2) = \{T \in \mathcal{L}(\mathbb{R}^2) : \det(T) = 1\}$

→ Claim: $\mathcal{L}^+(\mathbb{R}^2)$ is a subgroup of $\mathcal{L}(\mathbb{R}^2)$

Pf: Ex.

Q: What is the matrix group corresponding to $\mathcal{L}(\mathbb{R}^2)$?

A: $\mathcal{L}(2, \mathbb{R}) = \{M \in \mathcal{M}(2, \mathbb{R}) : M^T C M = C\}$, $C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\left[\mathcal{L}(4, \mathbb{R}) = \{M \in \mathcal{M}(4, \mathbb{R}) : M^T C M = C\}, \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \right]$$

→ Pf: 1) We show $\mathcal{L}(2, \mathbb{R})$ corresponds to $\mathcal{L}(\mathbb{R}^2)$.

Let $S: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ s.t. $S(v) = Mv \ \forall v \in \mathbb{R}^2$

$$S(v) \cdot S(w) = (Mv) \cdot (Mw) = (Mv)^T C (Mw) = v^T (M^T C M) w$$

Therefore $S \in \mathcal{L}(\mathbb{R}^2) \Leftrightarrow M^T C M = C$

2) We know that $L(2, \mathbb{R})$ is a group:

- $I \in L(2, \mathbb{R})$

- $M_1, M_2 \in L(2, \mathbb{R}) \Rightarrow (M_2 M_1)^T C (M_2 M_1) = \underbrace{M_1^T}_{M_1 \in L(2, \mathbb{R})} \underbrace{(M_2^T C M_2)}_{M_2 \in L(2, \mathbb{R})} M_1 = M_1^T C M_1 = C$

- Let $M \in L(2, \mathbb{R})$. Then

$$M^T C M = C \Rightarrow (\det M)^2 \cdot \underbrace{\det C}_{=-1} = \det C$$

$$\Rightarrow \det M = \pm 1 \neq 0$$

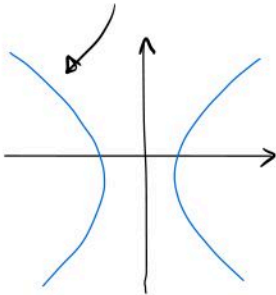
$$\Rightarrow \exists M^{-1} : M M^{-1} = M^{-1} M = I$$

$$\Rightarrow (M^{-1})^T M^T C M M^{-1} = (M^{-1})^T C M^{-1}$$

$$\Rightarrow \underbrace{(M M^{-1})^T}_{=I} C \underbrace{M M^{-1}}_{=I} = (M^{-1})^T C M^{-1}$$

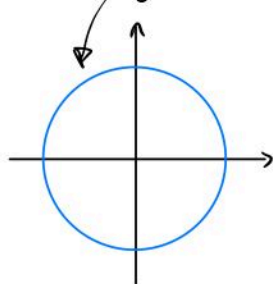
Def: $\sinh(\alpha) = \frac{e^\alpha - e^{-\alpha}}{2}$, $\cosh(\alpha) = \frac{e^\alpha + e^{-\alpha}}{2}$, hyperbolic trigonometric functions
 $\alpha \in \mathbb{R}$

$\rightarrow \forall (x, y) : x^2 - y^2 = 1 \Rightarrow \exists \alpha \in \mathbb{R} : (x, y) = (\cosh \alpha, \sinh \alpha)$



$\rightarrow$ Compare: $\sin(\alpha)$, $\cos(\alpha)$, $\alpha \in \mathbb{R}$ circular trigonometric functions

$\forall (x, y) : x^2 + y^2 = 1 \Rightarrow \exists \alpha \in \mathbb{R} : (x, y) = (\cos \alpha, \sin \alpha)$



Claim: $\mathcal{L}(2, \mathbb{R}) = \left\{ \begin{array}{l} \underbrace{\begin{pmatrix} \cosh(\alpha) & \sinh(\alpha) \\ \sinh(\alpha) & \cosh(\alpha) \end{pmatrix}}_{\text{rotation of 1st kind}}, \underbrace{\begin{pmatrix} \cosh(\alpha) & -\sinh(\alpha) \\ \sinh(\alpha) & -\cosh(\alpha) \end{pmatrix}}_{\text{reflection of 1st kind}} \\ \underbrace{\begin{pmatrix} -\cosh(\alpha) & \sinh(\alpha) \\ \sinh(\alpha) & -\cosh(\alpha) \end{pmatrix}}_{\text{rotation of 2nd kind}}, \underbrace{\begin{pmatrix} -\cosh(\alpha) & -\sinh(\alpha) \\ \sinh(\alpha) & \cosh(\alpha) \end{pmatrix}}_{\text{reflection of 2nd kind}} \end{array} \right\} \quad \alpha \in \mathbb{R}$

$\mathcal{L}^+(2, \mathbb{R})$

Pf: Let $M \in \mathcal{L}(2, \mathbb{R})$ and let T be the corresponding linear transformation. Let $T(1,0) = (a,b)$, $T(0,1) = (c,d)$

Assume $a > 0$.

- $T(1,0) \cdot T(1,0) = (1,0) \cdot (1,0) = 1 \Rightarrow a^2 - b^2 = 1$

$\Rightarrow \exists \alpha \in \mathbb{R}: T(1,0) = (\cosh(\alpha), \sinh(\alpha))$

- $T(1,0) \cdot T(0,1) = (1,0) \cdot (0,1) = 0 \Rightarrow ac - bd = 0$

$\Rightarrow c = \frac{bd}{a} = \frac{\sinh(\alpha)}{\cosh(\alpha)} d$

- $T(0,1) \cdot T(0,1) = (0,1) \cdot (0,1) = -1 \Rightarrow c^2 - d^2 = -1$

$\Rightarrow \left[1 - \left(\frac{\sinh(\alpha)}{\cosh(\alpha)} \right)^2 \right] d^2 = 1$

$\Rightarrow \frac{\cosh^2(\alpha) - \sinh^2(\alpha)}{\cosh^2(\alpha)} d^2 = 1$

$\Rightarrow d^2 = \cosh^2(\alpha) \Rightarrow d = \pm \cosh(\alpha)$

$\Rightarrow T(0,1) = \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \sinh(\alpha) \\ \cosh(\alpha) \end{pmatrix} \text{ or } T(0,1) = \begin{pmatrix} -\sinh(\alpha) \\ -\cosh(\alpha) \end{pmatrix}$

Assume $a < 0$: Ex.

Claim: $\mathcal{L}^+(2, \mathbb{R})$ is a subgroup of $\mathcal{L}(2, \mathbb{R})$

$\mathcal{L}^+(2, \mathbb{R})$ is a subgroup of $\mathcal{L}(2, \mathbb{R})$

Pf: Ex.

⑤ Generalization: $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ linear transformation

$v \cdot w$ nondegenerate symmetric bilinear form

$$v \cdot w = v^T C w$$

$$\text{Then } v \cdot w = T(v) \cdot T(w) \iff M^T C M = C$$

$\rightarrow O(\mathbb{R}^n)$ general orthogonal group

Remark: - If $C = M^T C M$ is anti-symmetric ($C^T = -C$), then

$Sp(n, \mathbb{R})$ is the symplectic group (classical Lie group)

- By Sylvester's theorem, every finite-dim. real vector space with nondegenerate symmetric bilinear form has a basis s.t.

$$C = \text{diag}(\pm 1)$$

$$r := \#(+1) = \text{signature of the form}$$